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B. E. (Fourth Semester) Examination, Nov.-Dec., 2015

(New Scheme)

(Elect. Engg. Branch)

ELECTRO MAGNETIC THEORY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d) of each question.

1. (a) Give two examples of vector and scalar fields. 2

(b) A uniform live change density of 5 nc/m is at $y = 0$,

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$z = 2$ m in free space, while -5 nc/m is located at $y = 0, z = -2$ m. A uniform surface charge density of $0{\cdot}3$ nc/m^2 at $y = 0{\cdot}2$ m and $-0{\cdot}3$ nc/m^2 is at $y = -0{\cdot}2$ m. Find $|\bar{E}|$ at the origin. 7

(c) Give the field

$$\bar{D} = \frac{20}{\rho^2}\left(-\sin\phi\,\overline{a\rho} + \sin 2\phi\,\overline{a\phi}\right) \text{c/m}^2$$

Find the total charge lying within the volume $1 < \rho < 2,\ 0 < \phi < \pi/2$ and $0 < z < 1$ by evaluate both side of divergence theorem. 7

(d) Given the points $P(\rho = 5, \phi = 60°, z = 2)$ and $Q\ (\rho = z, \phi = 110°, z = -1)$ in cylindrical coordinate system. Find :

(i) The distance $|\bar{R}_{PQ}|$

(ii) Unit vector in cartesian coordinates at P directed towards Q.

(iii) Unit vector in cylindrical coordinates at P directed towards Q.

2. (a) Discuss the Gauss law and its application. 2

(b) Within the cylinder $\rho = 2, 0 < z < 1$, the potential is

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given by $V = 100 + 50\rho + 150\rho \sin\phi$ find

$V, \bar{E}, \bar{D}$ and ρ_v at $P(1, 60°, 0{\cdot}5)$ in free space. 7

(c) Derive an expression for energy density in an electrostatic field. 7

(d) Derive expression for voltage and electric field due to an electric dipole. 7

3. (a) What is uniqueness theorem? 2

(b) The region $y < 0$ contains a dielectric material for which $\varepsilon r_1 = 2{\cdot}5$, while the region $\gamma > 0$ is characterized by $\varepsilon r_2 = 4$. Let

$$\bar{E}_1 = -30\,\overline{a_x} + 50\,\overline{a_y} + 70\,\overline{a_z} \text{ V/M}$$ 7

find :

(i) DN_2

(ii) EN_2

(iii) Dt_2

(iv) $\bar{D}_2$

(v) $\bar{P}_2$

(vi) θ_1

(vii) θ_2

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(c) Derive the potential field $V = \dfrac{50 \sin\theta}{r^2}$ V in free space (i) determine whether V satisfied Laplace's equation. (ii) Find the total charge stored inside the spherical shell $1 < r < 2$. 7

(d) (i) Derive continuity equation for current and define relaxation time. 4

(ii) Write short notes on method of image. 3

4. (a) Write any two difference between electrostatic potential and magnetic potential. 2

(b) Derive magnetic boundary condition and refraction law for magnetic flux. 7

(c) The magnetic field intensity is given in a certain region of space as 7

$$\bar{H} = \frac{(x + 2y)}{z^2}\,\overline{a_y} + \frac{2}{z}\,\overline{a_z} \text{ A/m}$$

(i) Find $\nabla \times \bar{H}$

(ii) Find $\bar{J}$

(iii) Use $\bar{J}$ to find the total current passing through

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the surface $z = 4, 1 \le x \le 2,\ 3 \le y \le 5$ in the $\overline{a}_z$ direction.

(iv) Show that the same result is obtained using other side of Stoke's theorem. 7

(d) (i) Define magnetic scalar and vector potentials. 4

(ii) State Biot-Savart's low for static magnetic field. 3

5. (a) What is displacement current density? 2

(b) Derive Maxwell's second equation from Ampere's circuital law in point form and integral form. 7

(c) Using Maxwell's equations. Derive Poynting's theorem and give its physical significance. 7

(d) Write down the Maxwell's equation in tabular form : 7

(i) General set

(ii) For static condition

(iii) For free static condition

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